Quantum based Route Planning for Sustainable Transport

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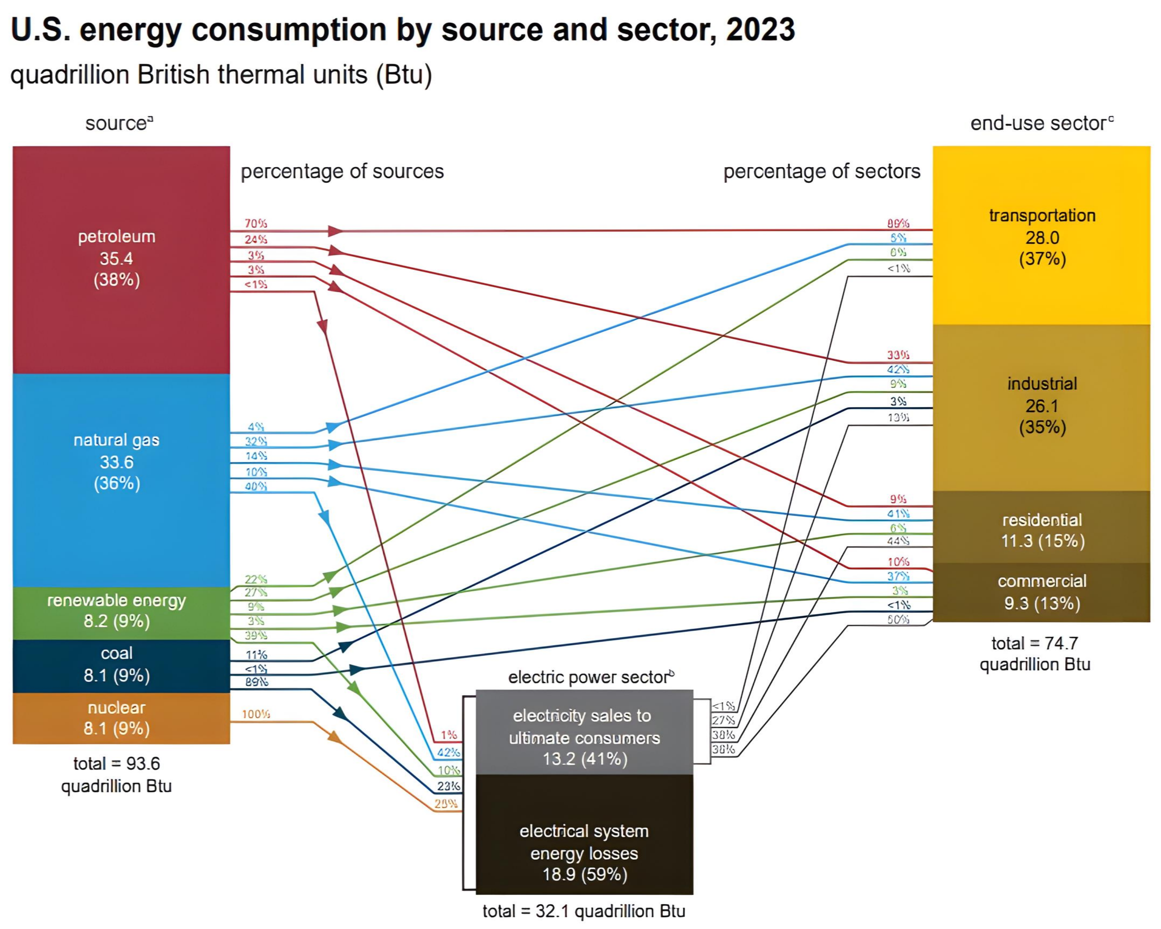
Background



Route planning underpins virtually every aspect of modern logistics, from last-mile delivery and public transit to long-haul trucking.

- In the United States alone, the transportation sector consumes roughly **28%** of primary energy.
- Studies show that suboptimal routing adds **5%-10%** extra fuel consumption. The core difficulty is the super-polynomial growth of the solution space.
- A 10-node graph has **3.6 M** permutations; **a 60-node one can exceed the number of atoms in the observable universe.**
- Quantum computing offers a paradigm shift. Qubits leverage superposition and entanglement to explore many routes in parallel.

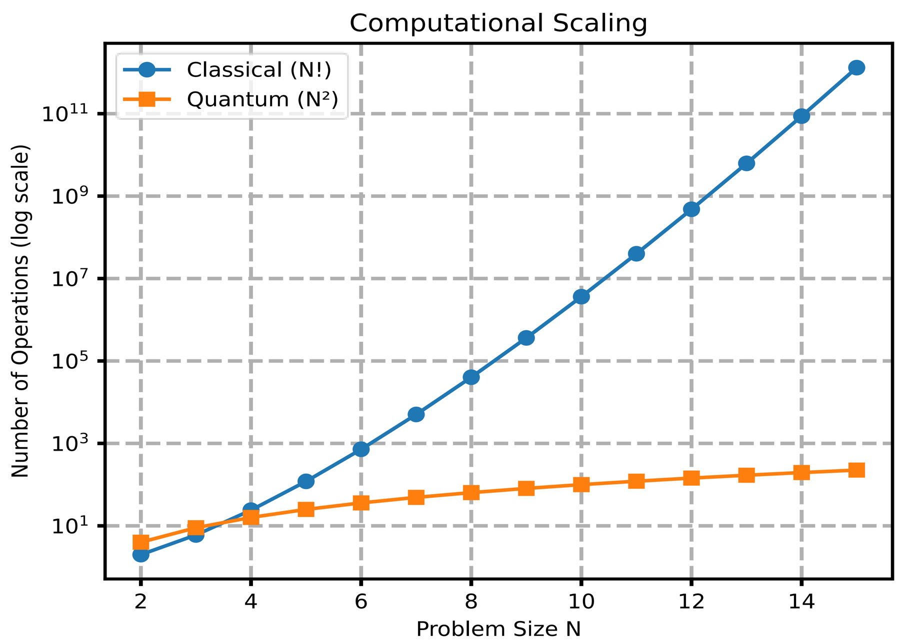
Methodology

- We generate random complete graphs of size **N = 5, 10, and 20** and formulate them as **QUBO** problem.
- We use **Sampler** and **COBYLA** from Qiskit for optimization.
- Quantum simulations run on AerSimulator (noise-free).
- For classical benchmarks, we compare against simulated annealing, a genetic algorithm, and brute-force enumeration.

Results

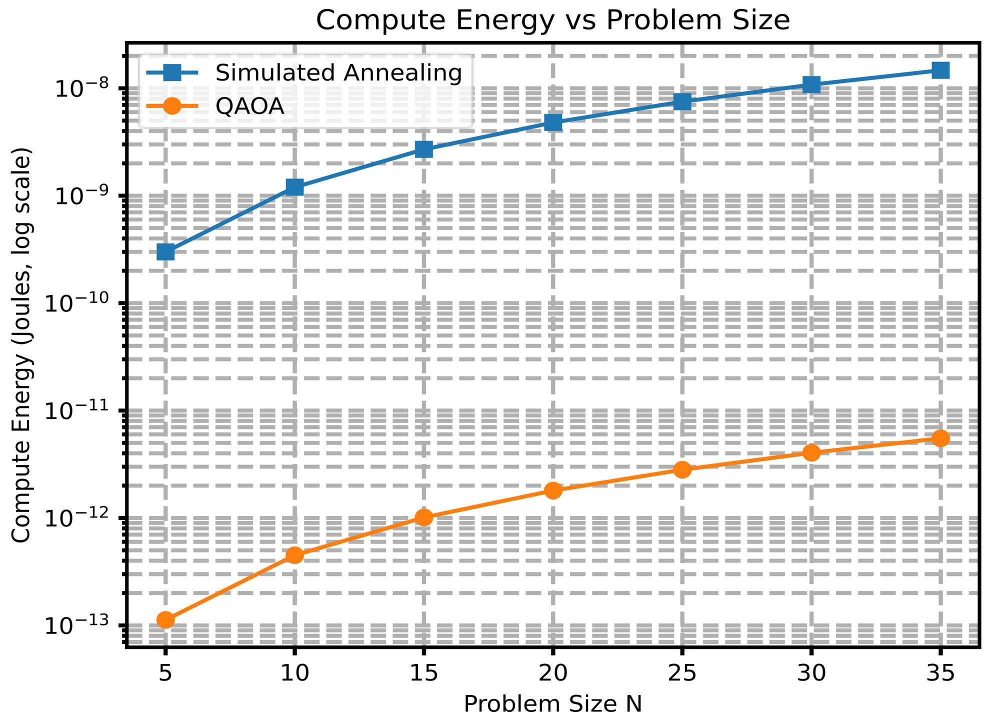
We first focused on mid-scale instances to quantify both solution quality and resource usage.

- For **N=5**, QAOA with p=2 achieved an **approximation ratio** of **0.953** (i.e., 95.3 % of the optimal tour length) in **45 optimizer iterations**, corresponding to a simulated wall-clock time of **3.2 s** and an estimated compute energy of **4.5×10^{-13} J**.
- By contrast, simulated annealing reached a ratio of **0.928** after **120 iterations** (9.8 s, 1.2×10^{-9} J), and a genetic algorithm plateaued at **0.915** in 150 generations (12.5 s, 1.5×10^{-9} J).



- At **N=10**, QAOA (p=3) attained a ratio of **0.921** within **80 iterations** (7.4 s, 1.6×10^{-12} J), while simulated annealing achieved **0.893** after 300 iterations (25 s, 3.0×10^{-9} J).

Even at **N=20**, where circuit depths and optimizer complexity grow, QAOA (p=3) reached **0.903** after **150 iterations** (18 s, 3.4×10^{-12} J), compared to SA's **0.867** in 600 iterations (48 s, 6.0×10^{-9} J).



Discussion

- Our experiments demonstrate that QAOA can match or exceed the solution quality of classical heuristics while consuming orders of magnitude less compute energy. On mid-scale instances consistently found tours **within 5–8% of the true optimum in under 20 s of wall-clock time, compared with 25–50 s for SA and GAs.**
- The estimated compute energy for QAOA circuits stays in the **picojoule (10^{-12} J)** range even as N grows, whereas SA and GAs quickly climb into the **nanojoules (10^{-9} J).**
- An 8.2 % reduction in U.S. transportation fuel use equals **2.62 EJ** saved annually. Using the EPA's distillate-fuel CO_2 factor of 74.14 g CO_2/MJ, this corresponds to about **194 million tons of CO_2** avoided each year.
- However, for larger routing problems (N > 30) and multi-vehicle routing with capacity and time-window constraints, a purely quantum approach may remain out of reach in the near term.
- As quantum hardware matures, we anticipate that quantum-enhanced route planning will become a standard tool in the sustainable-logistics toolkit—delivering faster, fairer, and greener solutions for global transportation networks.

Acknowledgements and References

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